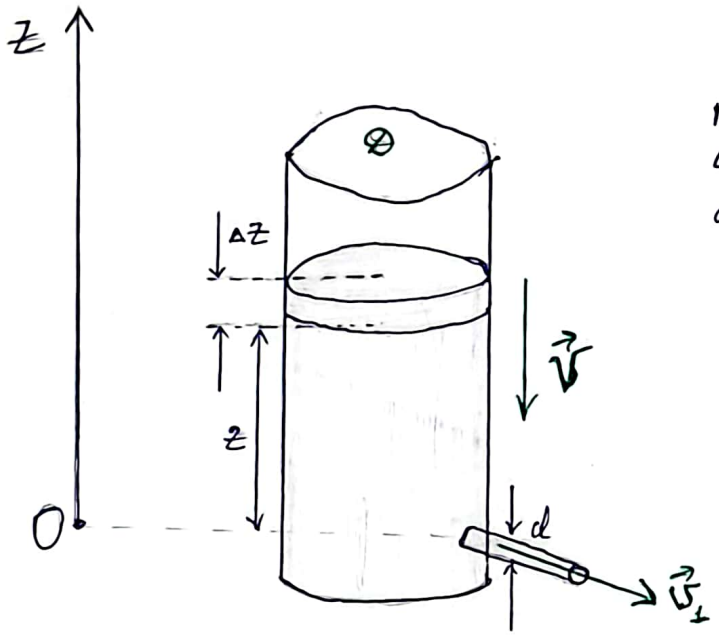


$\eta = \frac{d}{D} << 1.$

$d << z.$

❗ Masala boshqa usullarda ishlansa ham baholanadi.

a). $\Delta U = \Delta m g z$

$\Delta E_k = \frac{\Delta m v_1^2}{2}$

$\frac{\Delta m v_1^2}{2} = \Delta m g z$ (1 ball)

$v_1 = \sqrt{2gz}$ (1 ball)

b) $S v \Delta t = S_1 v_1 \Delta t$ (1 ball)

$v = \frac{S_1}{S} v_1 = \left(\frac{d}{D}\right)^2 v_1 = \eta^2 \sqrt{2gz}$ (0,5 ball)

$v_z(z) = -\sqrt{2(\eta^4 g)z}$ (0,5 ball)

c). Tezlanish topishning fizik ma'nosini so'z bilan yozgan bo'lsa (0,5 ball)
matematik (formulalardan) keltirib chiqarsa (0,5 ball)

$a = +\eta^4 g$ (1 ball)

d). $z(t) = \frac{(v_0 - \eta^4 g t)^2}{2\eta^4 g}$ (1 ball)

$z(t) = \frac{(\sqrt{2\eta^4 g h_0} - \eta^4 g t)^2}{2\eta^4 g}$ (1 ball)

e) $\tau_{0,5} = (\sqrt{2} - 1)\sqrt{\frac{h_0}{\eta^4 g}}$ (1 ball)

f) $\tau_{0,5} = (\sqrt{2} - 1) \cdot 20^2 \cdot \sqrt{\frac{0,20}{10}} \approx 23,43\ s$ (1 ball)

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